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PI Name:	Kassemi, Mohammad Ph.D.		
Project Title:	The Zero Boil-Off Tank-NonCondensibles (ZBOT-NC) Experiment		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Comments:	NOTE (Dec 2019): former affiliation included National Center for Space Exploration Research (NCSER), per information from J. McQuillen/GRC		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Assignment:	NOTE: End date changed to 05/31/2026 per the PI (Ed., 2/13/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Meyer, Michael L (NASA Langley Research Center) Koochesfahani, Manoochehr (Michigan State University)		
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Task Description:

Principal Investigator (PI): Prof. Mohammad Kassemi (Case Western Reserve University/NASA Glenn Research Center (GRC)); Co-Investigator: Michael L. Meyer (NASA Langely); Co-Investigator: Prof. Manoocher Koochesfahani (Michigan State University) Project Scientist: Dr. Howard Pearlman (NASA GRC); Project Manager: Daniel F. Brown (NASA GRC).
Synopsis NASA's projected exploration program includes a series of human and robotic expeditions to low and high earth orbit, Moon, Mars, and possibly the asteroids and moons of other planets. Integral to all phases of these space and planetary expeditions is affordable and reliable cryogenic fluid storage for use in propellant or life support systems. Without safe, efficient, and flexible cryogen storage, economically justifiable human missions may not be possible. Cryogen vaporization caused by various heat leaks into a tank from its surroundings is the main cause of storage tank self-pressurization. Without dynamic and active pressure control, self-pressurization can only be countered by continuous venting of the vapor to preserve the tank's structural integrity. This leads to unacceptable fuel loss levels for long-duration missions and has given impetus to the development of innovative zero-boil-off or reduced-boil-off pressure control designs based on mixing of the bulk fluids with or without active cooling. Unfortunately, both tank self -pressurization and pressure control are governed by intricate and complicated dynamic interactions among forced mixing, various gravity-dependent transport mechanisms in the vapor and liquid phases, and the multifaceted condensation/evaporation processes at the interface. Consequently, effective implementation and optimization of a dynamic pressure control system for space applications cannot be accomplished empirically, especially since there is a serious scarcity of relevant microgravity data.

In order to address this need, and in line with the strong recommendations provided by the National Academy of Sciences' (NAS) National Research Council (NRC) Decadal Survey, the ZBOT series of hierarchical microgravity science experiments were proposed and initiated with the following overall goals: 1. ZBOT-1–FM (fluid mixing): Investigate transport and phase change phenomena associated with self-pressurization & zero-boil-off (ZBO) mixing-jet pressure control in a sealed tank, in microgravity. Lay the foundation for the development and validation of a state-of-the-art two-phase Computational Fluid Dynamics (CFD) model for storage tank cryogenic fluid management (CFM). 2. ZBOT-NC (noncondensable effects): Investigate the flow, transport, and kinetics effects of noncondensable gas (NCG) pressurants and their impact on the interfacial condensation and evaporation processes during tank self-pressurization and pressure control in microgravity. To be accompanied also with hand-in-hand CFD model development and validation. 3. ZBOT-AC (active cooling): Investigate the heat and mass transport and phase change characteristics of droplet injection using a spray bar during tank pressure control and chilldown operations in microgravity, together again, with hand-in-hand CFD model development and validation.

As a first step towards achieving these goals, the ZBOT-1 experiment was performed aboard the International Space Station (ISS) in 2017. The experiment met its Complete Scientific Success Criteria and proved that such a cost-effective, subscale, simulant fluid, ISS-class science experiment can provide valuable microgravity data for: • Successful and comprehensive validation and verification of state-of-the-art CFD models that will be ultimately used to design propellant tanks for future space missions. • Gaining a better understanding of the intricate transport and phase change interactions that control tank pressurization and pressure control in microgravity. • Pointing out several serious practical implications of two-phase phenomena that may prove important to future large cryogenic storage tank operations operating in microgravity.

The main goal of the present ZBOT-NC experiment is to determine the important but inadequately understood and assessed impact of non-condensable pressurant gases on the interfacial evaporation/condensation and bulk transport processes that affect tank pressurization and pressure control in microgravity. Building upon the success of ZBOT-1, the specific scientific objective in this second experimental–computational investigation will be to clarify the following phenomena: 1. The nature of the transport barrier created by the NCG in the ullage and its effect on interfacial evaporation and condensation rates in microgravity. 2. The effect of a non-condensable gas on the interfacial mass transfer kinetics, and the extent of any resistance to interfacial mass transfer caused by the gas penetrating the Knudsen layer 3. The existence, nature, and extent of a unique class of Marangoni (thermocapillary) convection in the liquid that is driven by partial pressure variations in the vapor. 4. Combined impact of the above on the global pressurization/stratification and pressure reduction/destratification rates in microgravity. This enhanced fundamental scientific knowledge and insight will be gained while we investigate the effect of noncondensable gases on: (a) tank thermal stratification, self-pressurization and wall boiling as a function of global and localized heat loads, tank fill levels, and the noncondensable gases' molecular weights and concentrations; and (b) thermal destratification and depressurization rates as a function of jet temperature, jet Weber numbers (velocities), tank fill levels, and the noncondensable gases' molecular weight and concentrations. In order to achieve these objectives, we plan on performing: • high accuracy measurement of the ullage pressure • high accuracy local temperature measurements at multiple locations in the tank interior and on the tank wall • accurate determination of NCG concentrations in the ullage • visualization of the liquid-vapor interface through white light imaging; and • non-intrusive capture of the liquid temperature fields using state-of-the-art Quantum Dot Thermometry (QTD) diagnostics that will be developed and customized in-house.

The local instrumentation/sensor data and the full-field image data will be used to validate a two-phase two-component CFD storage tank model that will be developed hand-in-hand with the experiment. This will be accomplished by building upon the solid foundation of model development and validation that was initiated and carried out quite successfully during the ZBOT-1 investigation.

It is anticipated that the acquisition of the relevant microgravity data and the development of the much-needed microgravity empirical correlations for mixing and destratification in the presence of noncondensable gases, together with development, verification, and validation of the CFD models and codes, that will be ultimately used in the scale-up design of the future storage tank prototype, will all contribute significantly to reduce the uncertainty, cost, and risk associated with the development of both cryogenic upper stages and in-Space or on-surface fuel depots for future NASA planetary missions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	The result of the ZBOT investigations will primarily and directly address the fundamental need of NASA and the aerospace industry for efficient, reliable, and enabling propellant and life support storage and transfer technologies for future long-duration planetary exploration. The fundamental two-phase data and scientific understanding provided by the experiment and the associated state-of-the-art two-phase theoretical model development will also aid in the improvement of a set of diverse ground-based two-phase technologies such as condensing heat exchangers, phase separators, water-handling operations in fuel cells, and storage and transport of cryogenic and non-cryogenic consumable fluids on Earth. In this light, it is strongly aligned with the global impetus to transition to a clean and green hydrogen economy vigorously pursued by Japan, Europe, and the US.
Task Progress:	The project has successfully built upon the strong foundation of the ZBOT-1 flight design, and through the diligent work of the highly experienced science and engineering teams, the project has made excellent progress on the hardware development, diagnostic development, CFD and Thermodynamic Model development, and breadboard testing fronts. [Ed. Note: For earlier project, see Kassemi / The Zero Boil-Off Tank (ZBOT) Experiment]. Two major breadboard development and testing tasks have been performed. The first was to characterize the two-phase flow loop with Perfluoro-N-Pentane (PNP) and the injected noncondensable gases (NCG): GN2, GNe, GXe. The second was to assess the performance of the innovative and novel Quantum Dot Thermometry technique that is being developed as part of the flight project for measuring the test cell full temperature field non-intrusively. Additionally, finite element modeling case studies were performed to minimize residual heat leak paths and enhance the uniformity of thermal boundary and initial conditions. Finally, the Stage-I sharp-interface Computational Fluid Dynamics (CFD) two-phase model for the two-component PnP-NCG system has been developed and verified through initial parametric testing. Several hardware redesigns that are being implemented will potentially provide marked improvements over ZBOT-1 hardware and testing performance through the imposition of: • more uniform thermal boundary and initial conditions • vapor free single-phase jet mixing • straight non-skewed inlet jet flows • reduced propensity for inside test cell cavitation • improved monitoring and control of heat leaks into the tank • improved control and maintenance of tank fill levels Moreover, comprehensive and detailed breadboard flow loop experiments under microgravity-simulating orientations have shown successful injection of noncondensable gases that will remain in solution at jet nozzle exit and in the vicinity of heat sources – thus removing two areas of prime concern for undesirable bubble evolution. Significant progress has been also made for improving the white light optical diagnostics and QDT: • Potentially significant enhancement of the white light image capture capability over ZBOT-1 is expected through innovative uses of: LED light spectrum; background lighting; and vacuum jacket coating to reduce light scattering. • Feasibility of QDT has been established through comprehensive scientific laboratory breadboard testing. The next crucial step is to test QDT performance under flight-like hardware constraints with the possible risk that it may not yield the same optimum science performance achieved under the relatively less constrained laboratory breadboard settings. Finally, great strides have been made in the development of the Stage-I sharp-interface Computational Fluid Dynamics (CFD) two-phase model for the two-component PnP-NCG system. Initial simulation case studies have shown the profound and significant impact of the noncondensable gas on tank pressure evolution, especially during jet mixing pressure control in microgravity.
Bibliography Type:	Description: (Last Updated: 07/29/2025)
Abstracts for Journals and Proceedings	Hylton S, Kassemi M, Kahwaji MJ, Magee K. "Experimental and CFD investigation of the tilted jet behavior in the Zero Boil-Off Tank (ZBOT) Experiment." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021